

Main Research Figures and Tables

Evaluating Recommender Systems under Sparse User Feedback

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1 Datasets and Models

Table 1: Combined dataset and preprocessing statistics. Positive feedback uses ratings ≥ 4 ; Amazon repeated positive user-item pairs retain the most recent interaction. The final two interactions per user are held out for validation and test.

Statistic	MovieLens-1M	Amazon Video Games
Raw – users	6,040	2,766,656
Raw – items	3,706	137,249
Raw – interactions	1,000,209	4,624,615
Positive feedback before deduplication – users	6,038	2,115,867
Positive feedback before deduplication – items	3,533	120,360
Positive feedback before deduplication – interactions	575,281	3,445,132
Positive feedback after deduplication – users	6,038	2,115,867
Positive feedback after deduplication – items	3,533	120,360
Positive feedback after deduplication – interactions	575,281	3,394,554
Iterative 5-core – users	6,034	63,413
Iterative 5-core – items	3,125	19,022
Iterative 5-core – interactions	574,376	533,550
Training interactions	562,308	406,724
Validation interactions	6,034	63,413
Test interactions	6,034	63,413
Mean history length before holdout	95.19	8.41
Mean training history length	93.19	6.41

Table 2: Evaluated recommender systems and model families.

Model	Family	Final runs per condition
Pop	Non-personalized	1
ItemKNN	Classical	1
BPR	Classical	3
EASE	Linear	1
NeuMF	Neural	3
MultiVAE	Neural	3
GRU4Rec	Sequential	3
SASRec	Sequential	3
BERT4Rec	Sequential	3
LightGCN	Graph	3

2 Sparsity and Data Conditions

Random, Recent and Early retain a random subset, the most recent interactions and the earliest interactions of each training history, respectively. Retained counts are $\max(1, \lceil nr \rceil)$ for training-history length n and nominal retention r . The post-preprocessing item catalogue remains fixed.

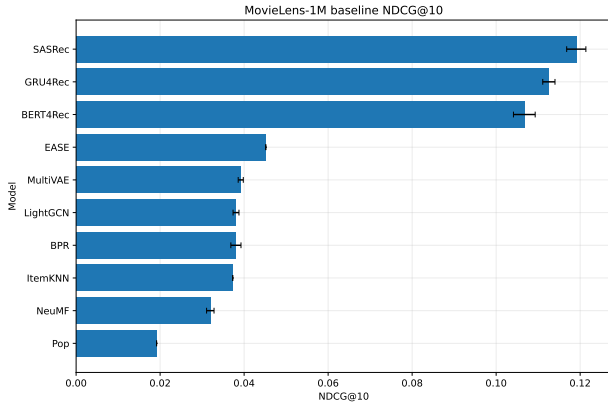
Table 3: Training retention and unseen test targets. Unseen percentages count test targets whose items occur nowhere in retained training data. Counts and mean history lengths are identical across scenarios at each retention level.

Dataset	Nominal	Actual	Mean/user	Unseen test targets		
				Random	Recent	Early
MovieLens	100%	100.000%	93.19	0.000%	0.000%	0.000%
MovieLens	50%	50.270%	46.85	0.033%	0.000%	0.099%
MovieLens	25%	25.404%	23.67	0.265%	0.066%	0.563%
MovieLens	10%	10.479%	9.76	1.425%	0.580%	2.320%
Amazon	100%	100.000%	6.41	0.358%	0.358%	0.358%
Amazon	50%	54.712%	3.51	1.667%	1.116%	3.331%
Amazon	25%	29.767%	1.91	4.889%	3.301%	8.621%
Amazon	10%	18.380%	1.18	9.071%	6.894%	13.177%

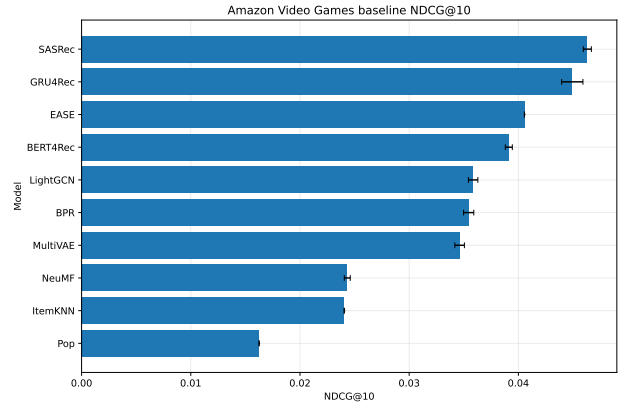
3 Baseline Results

Table 4: Cross-dataset baseline results. Values are means over the final runs listed in Table 2.

Model	MovieLens-1M			Amazon Video Games		
	NDCG@10	Recall@10	MRR@10	NDCG@10	Recall@10	MRR@10
SASRec	0.119079	0.223898	0.087307	0.046315	0.096300	0.031080
GRU4Rec	0.112553	0.210032	0.083042	0.044938	0.084983	0.032841
BERT4Rec	0.106708	0.212463	0.074836	0.039125	0.074291	0.028502
EASE	0.045156	0.090322	0.031692	0.040555	0.072666	0.030816
MultiVAE	0.039195	0.081151	0.026741	0.034618	0.064361	0.025627
LightGCN	0.038068	0.079604	0.025681	0.035857	0.065933	0.026761
BPR	0.038047	0.079218	0.025787	0.035450	0.065002	0.026504
ItemKNN	0.037286	0.075572	0.025903	0.024051	0.043934	0.018025
NeuMF	0.031950	0.067120	0.021448	0.024331	0.046247	0.017706
Pop	0.019158	0.039443	0.013127	0.016255	0.032012	0.011597



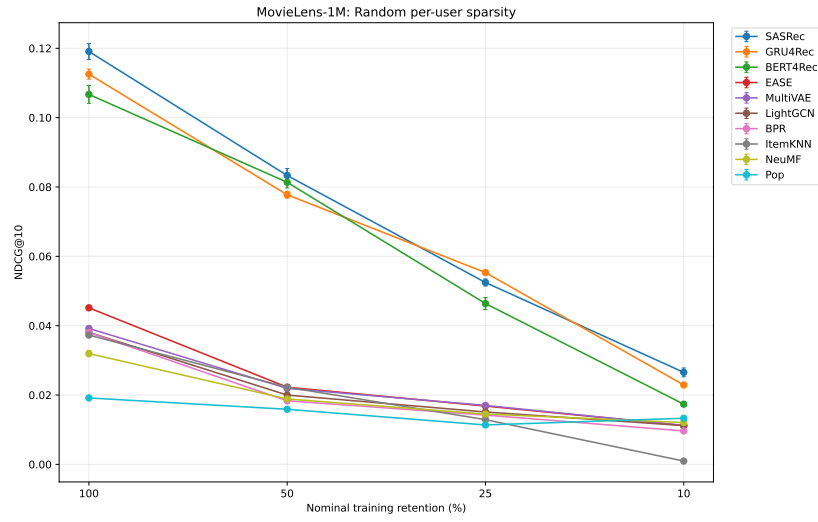
(a) MovieLens-1M.



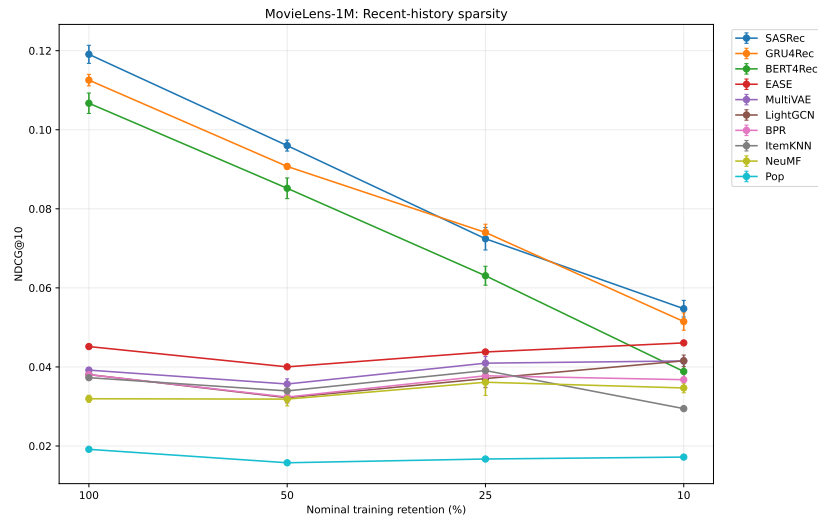
(b) Amazon Video Games.

Figure 1: Baseline NDCG@10 performance across all models. Error bars indicate one sample standard deviation for stochastic models.

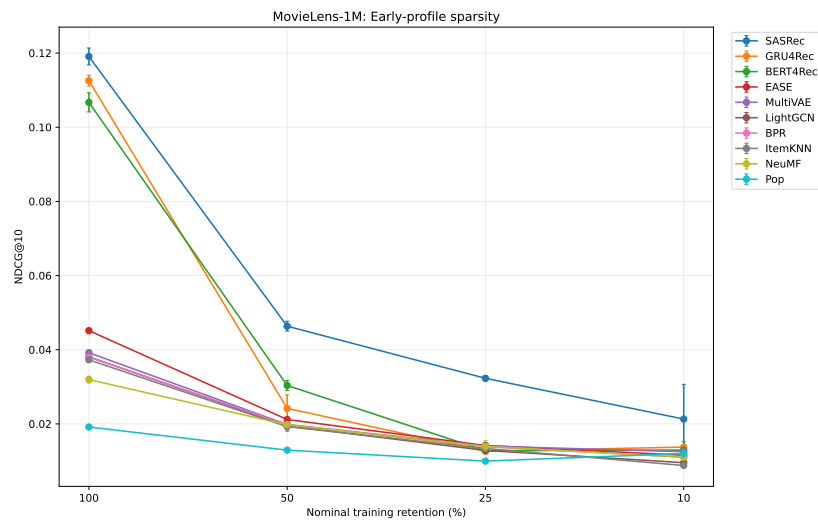
4 Sparsity Curves



(a) Random

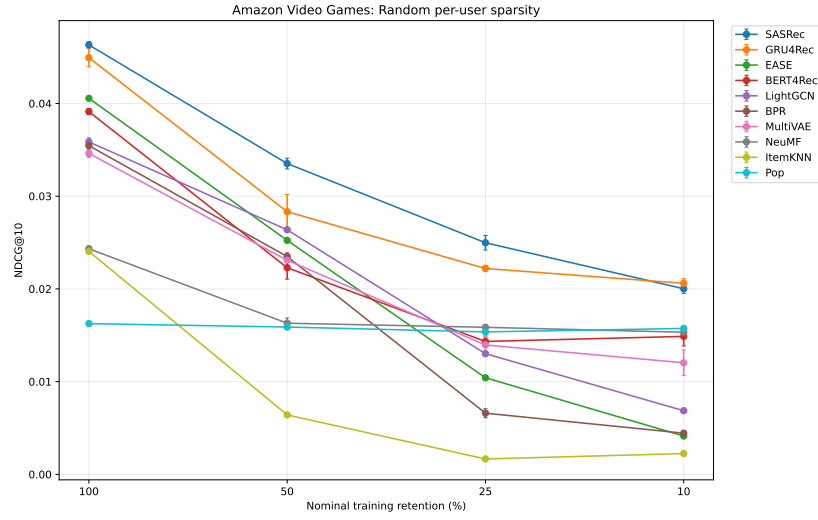


(b) Recent

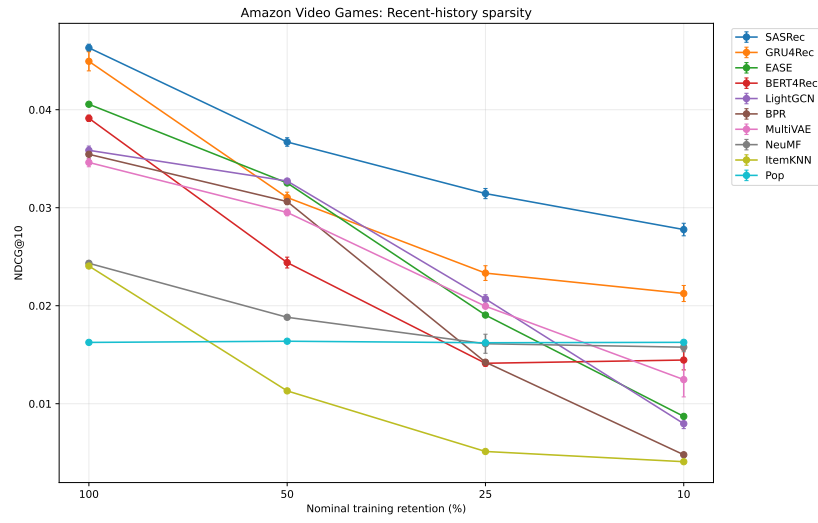


(c) Early

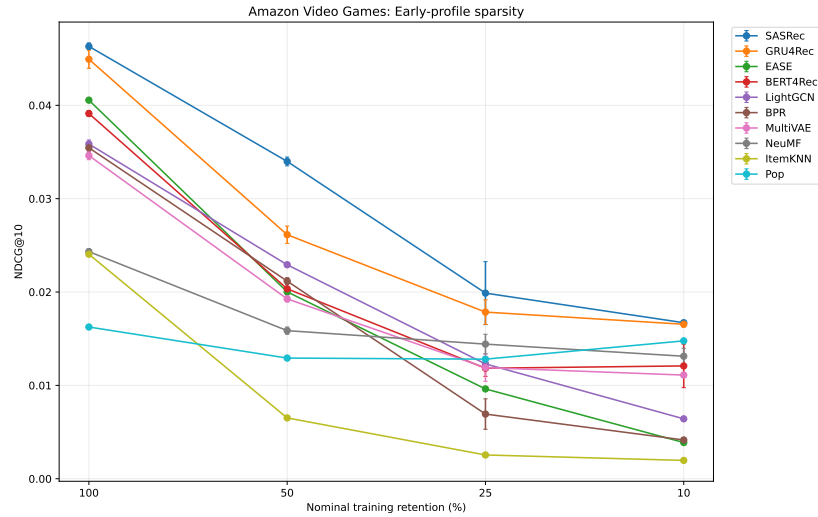
Figure 2: MovieLens-1M NDCG@10 as training-history retention decreases.



(a) Random



(b) Recent



(c) Early

Figure 3: Amazon Video Games NDCG@10 as training-history retention decreases.

5 Relative Changes and Ranking Movement

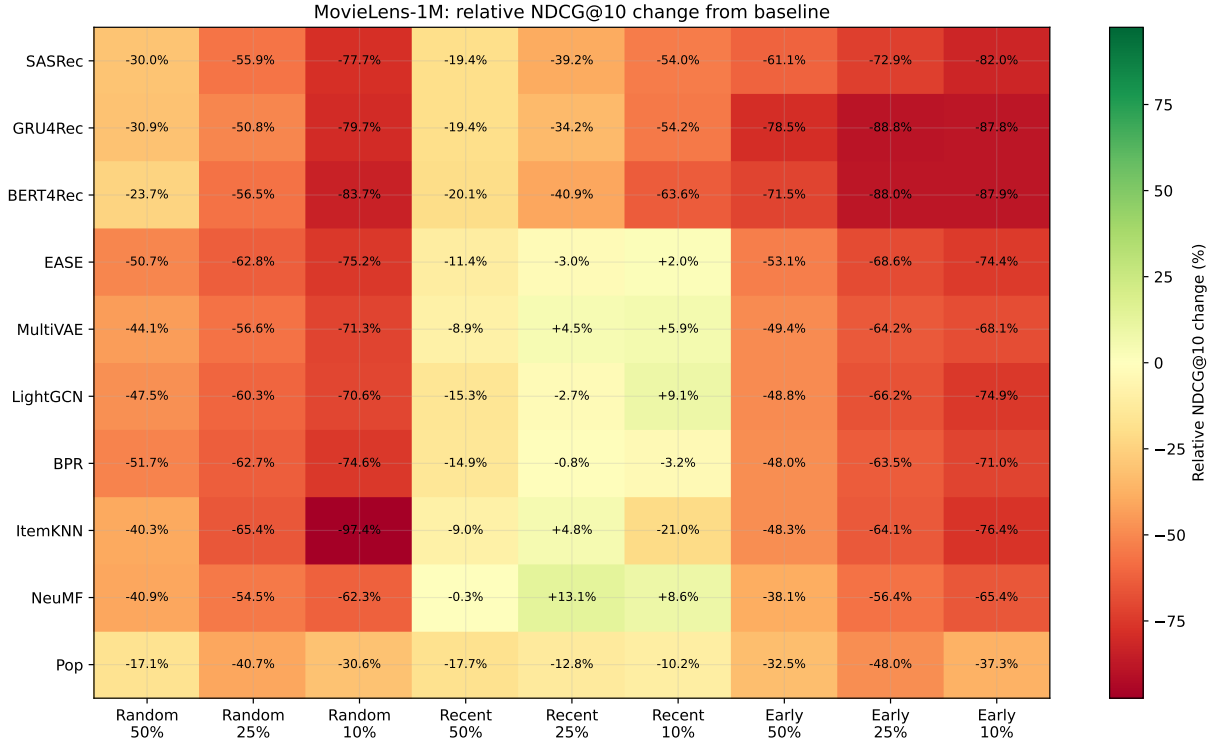


Figure 4: MovieLens relative NDCG@10 change from each model’s full-data baseline. Negative values indicate degradation; positive values indicate improvement.

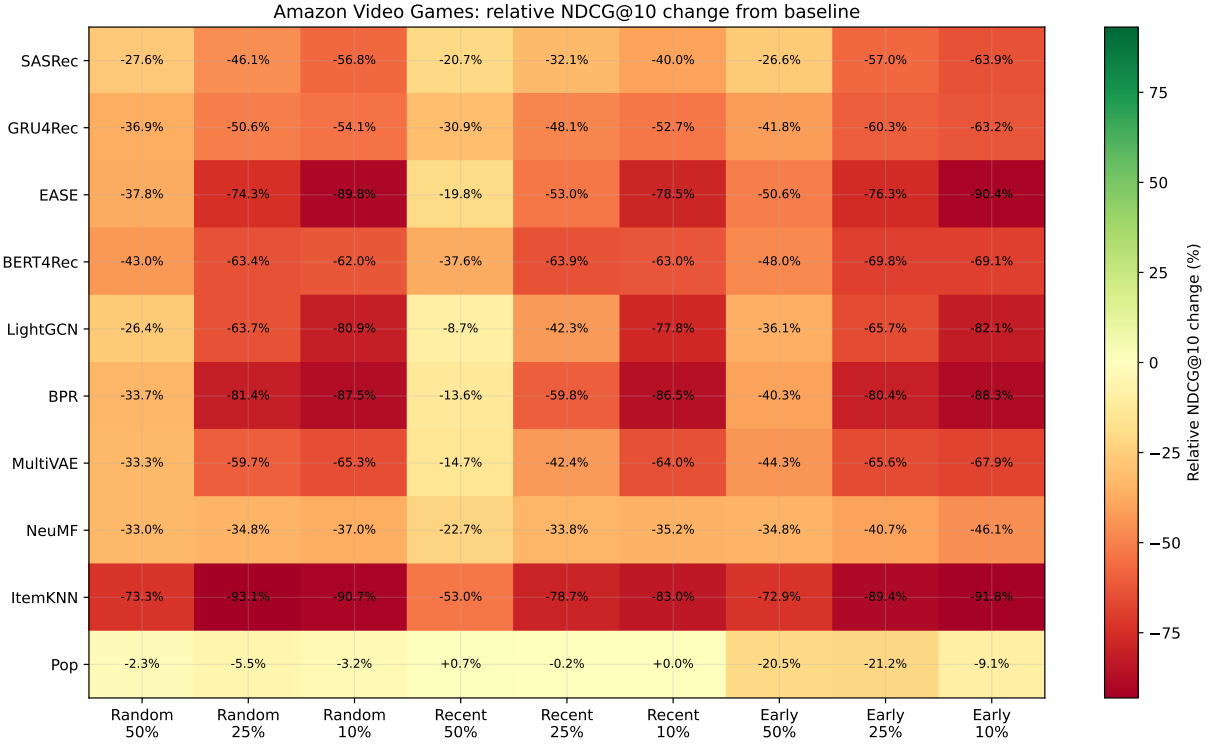


Figure 5: Amazon relative NDCG@10 change from each model’s full-data baseline. Negative values indicate degradation; positive values indicate improvement.

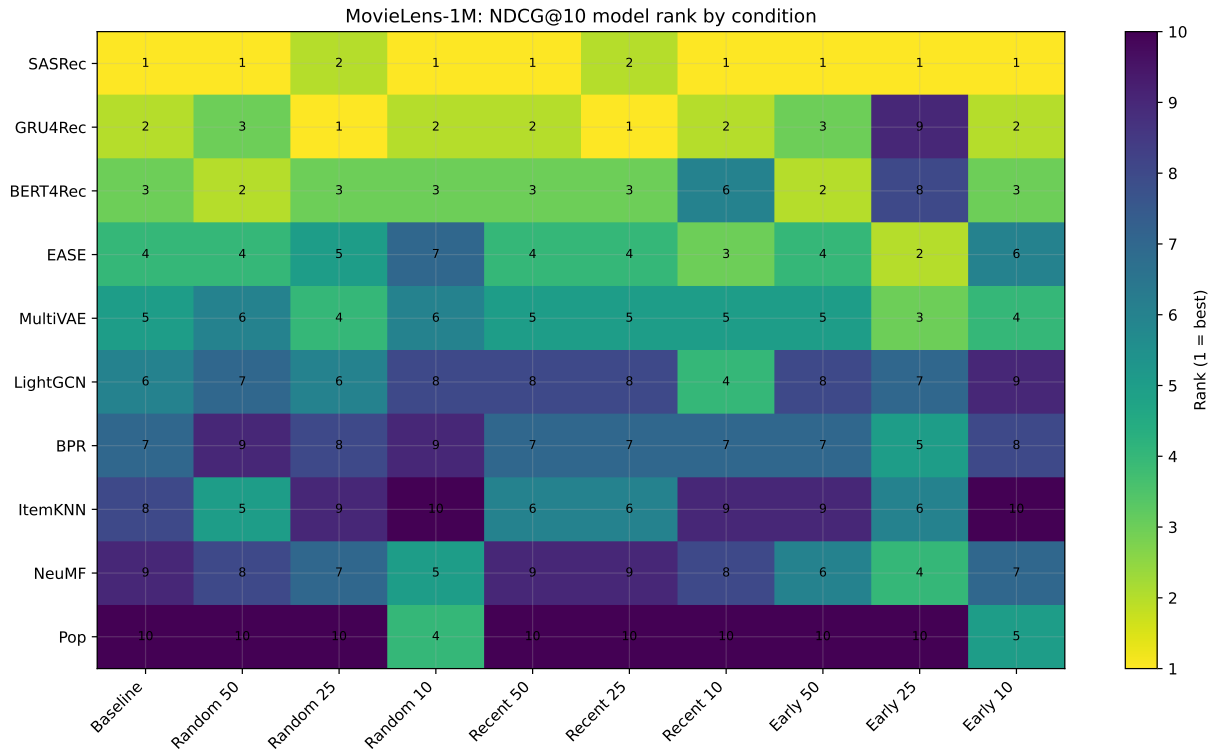


Figure 6: MovieLens NDCG@10 model ranks across baseline and sparsity conditions.

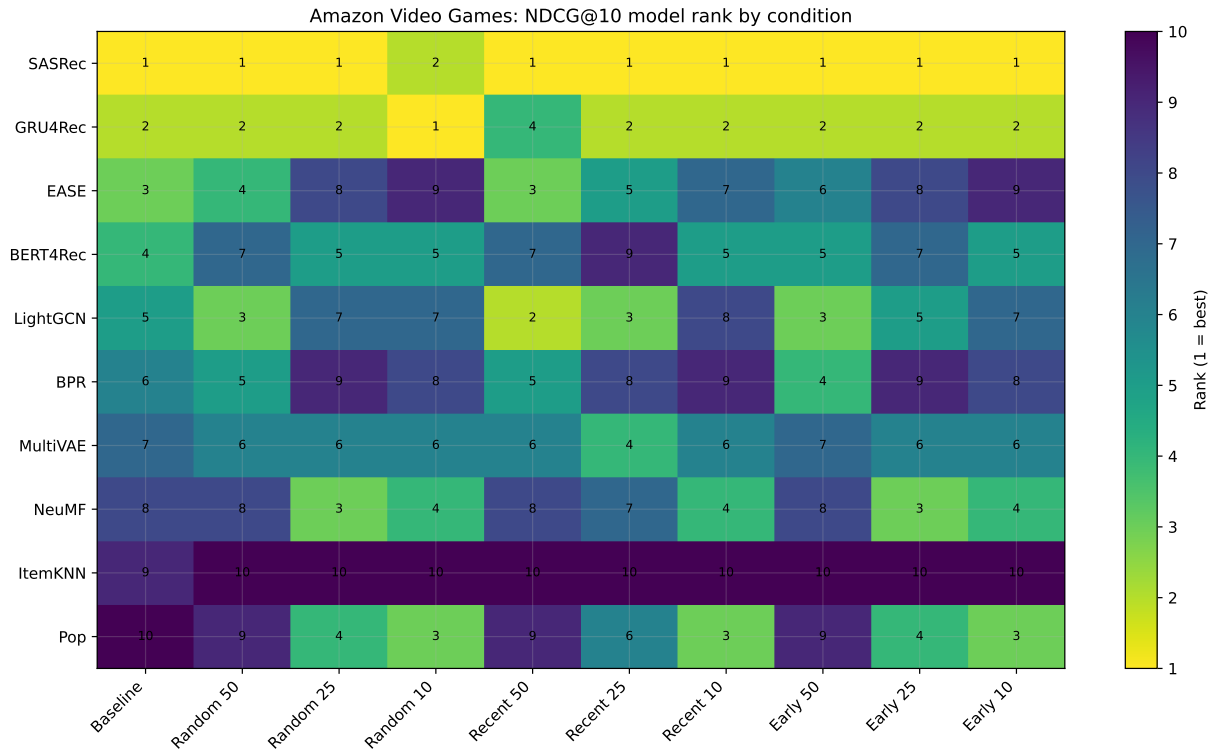


Figure 7: Amazon NDCG@10 model ranks across baseline and sparsity conditions.

6 Severe Sparsity and Median Relative Change

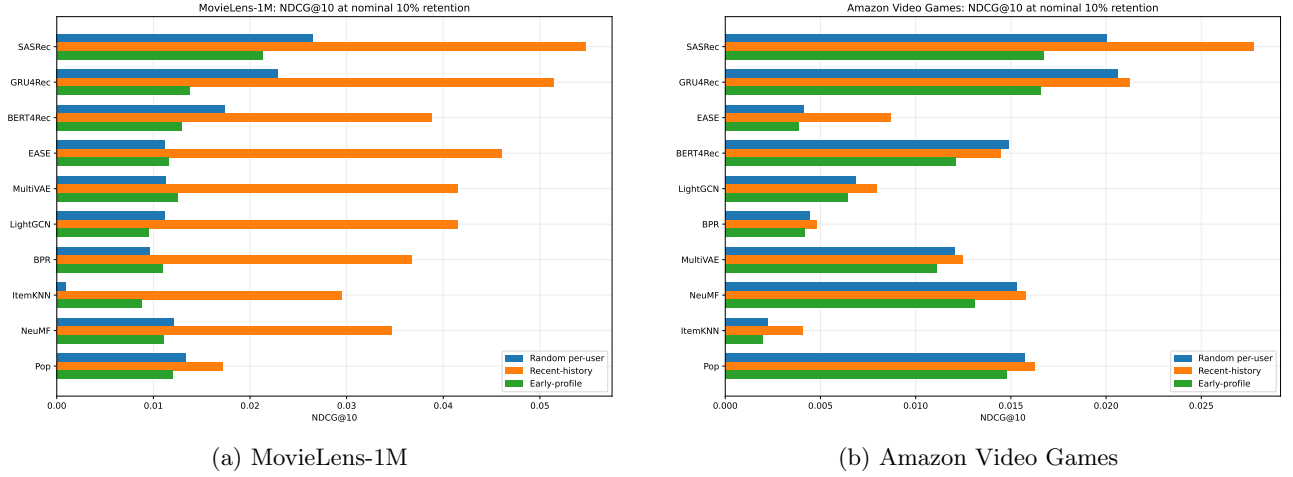


Figure 8: NDCG@10 under Random, Recent and Early sparsity at nominal 10% retention.

Table 5: Median relative NDCG@10 change across the ten models.

Scenario	Nominal retention	MovieLens	Amazon
Random	50%	-40.62%	-33.50%
Random	25%	-56.57%	-61.52%
Random	10%	-74.91%	-63.61%
Recent	50%	-15.10%	-20.27%
Recent	25%	-2.87%	-45.22%
Recent	10%	-6.73%	-63.54%
Early	50%	-49.08%	-41.02%
Early	25%	-65.24%	-65.64%
Early	10%	-74.68%	-68.52%

7 Unseen Targets and Condition Winners

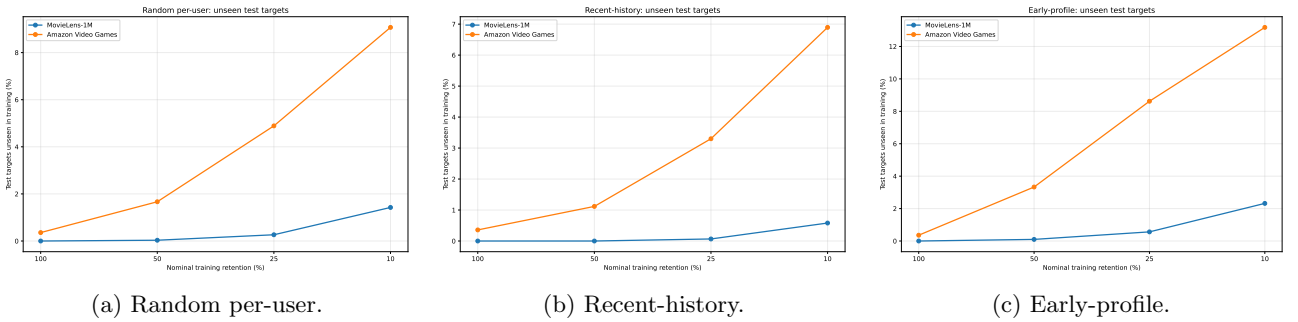


Figure 9: Percentage of held-out test targets whose items become globally unseen in retained training data.

Table 6: Highest mean NDCG@10 model in each experimental condition.

Condition	MovieLens-1M	Amazon Video Games
Baseline	SASRec	SASRec
Random 50%	SASRec	SASRec
Random 25%	GRU4Rec	SASRec
Random 10%	SASRec	GRU4Rec
Recent 50%	SASRec	SASRec
Recent 25%	GRU4Rec	SASRec
Recent 10%	SASRec	SASRec
Early 50%	SASRec	SASRec
Early 25%	SASRec	SASRec
Early 10%	SASRec	SASRec

SASRec wins 17 of the 20 dataset-condition combinations; GRU4Rec wins the remaining three.

8 Cross-Dataset Ranking and Robustness

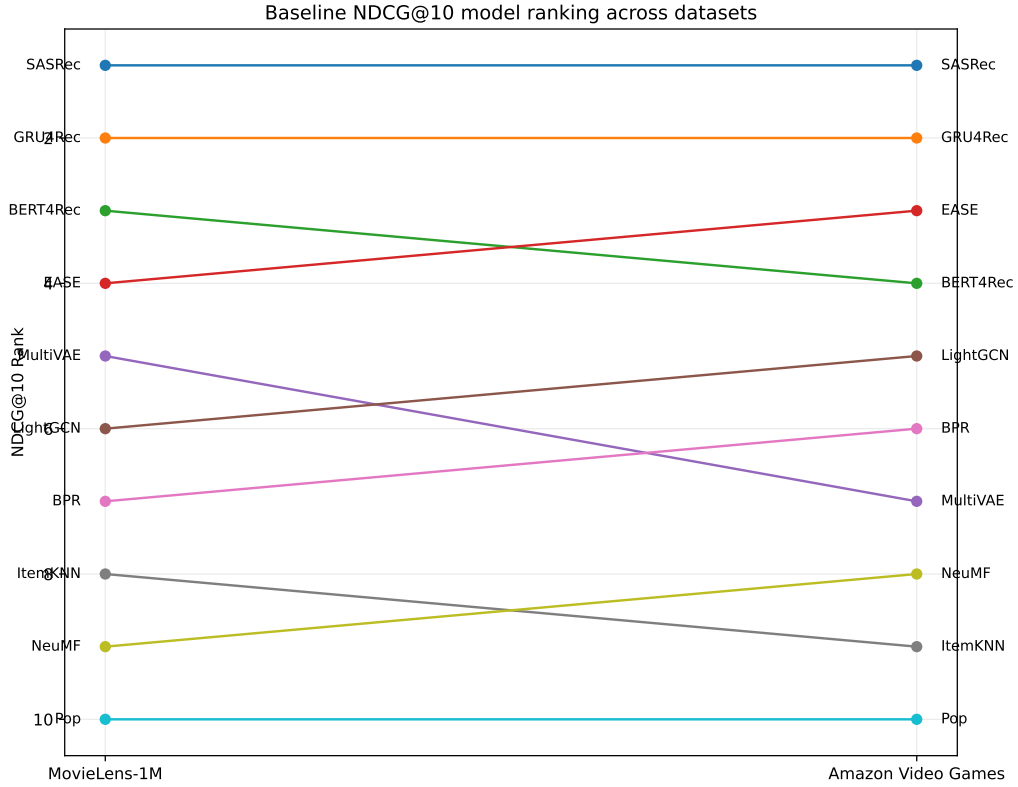


Figure 10: Baseline NDCG@10 model-rank movement between MovieLens-1M and Amazon Video Games.

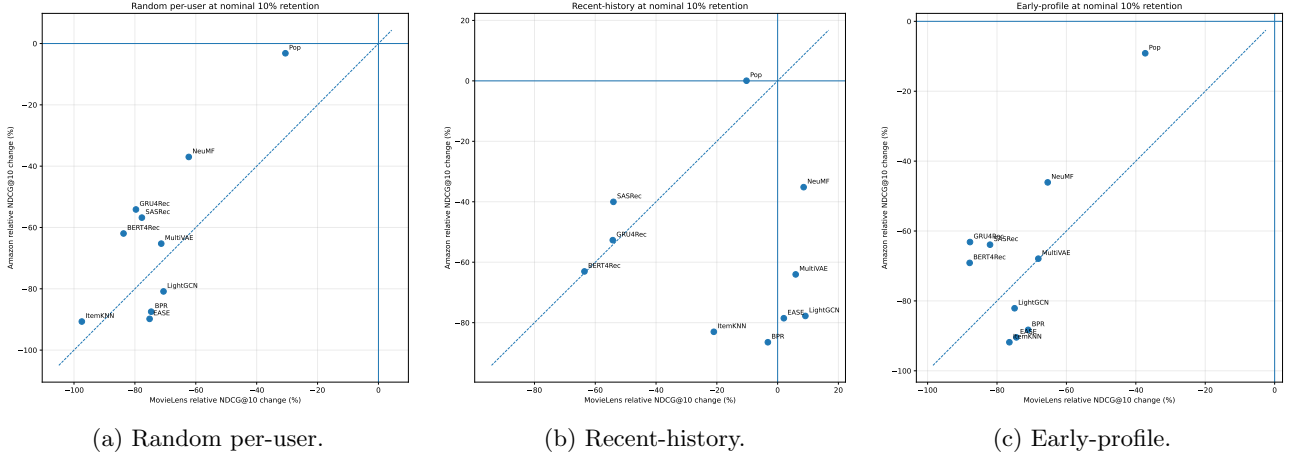


Figure 11: Model-level relative NDCG@10 change across datasets at nominal 10% retention. The diagonal represents equal relative change on the two datasets.

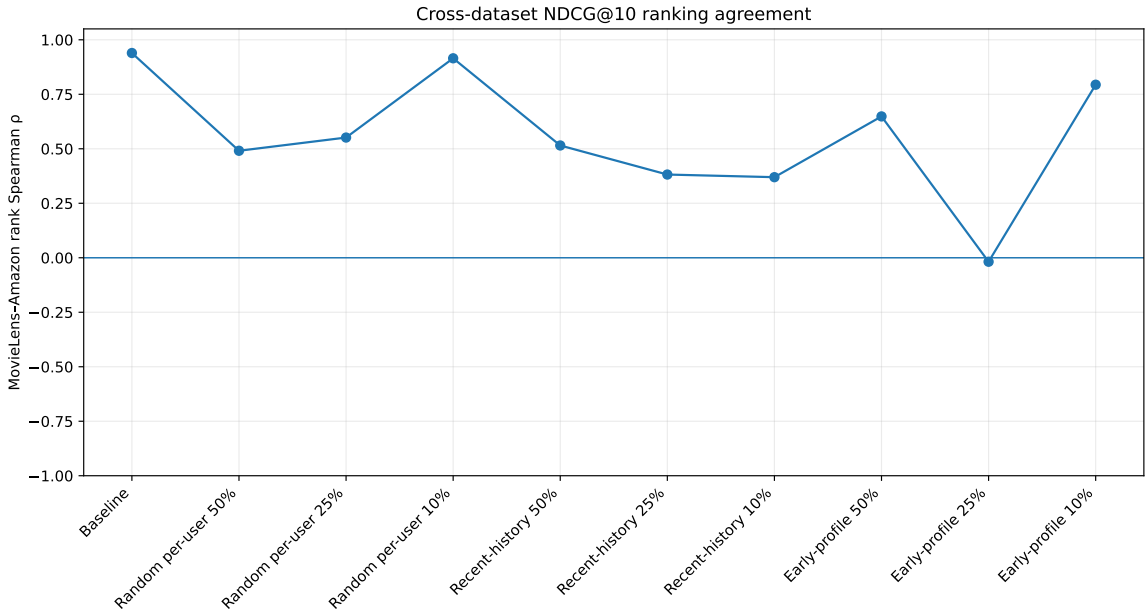


Figure 12: Spearman agreement between MovieLens and Amazon NDCG@10 model rankings across matching experimental conditions.

9 Metric and Recommendation-Length Consistency

Table 7: Spearman agreement with NDCG@10 model rankings: median (minimum–maximum) over baseline and nine sparse conditions. Summaries use the reported three-decimal correlations.

Compared metric	MovieLens-1M	Amazon Video Games
Recall@10	0.988 (0.927–0.988)	0.988 (0.952–1.000)
MRR@10	0.982 (0.867–1.000)	0.994 (0.927–1.000)
NDCG@5	0.988 (0.842–1.000)	0.988 (0.915–1.000)
NDCG@20	0.988 (0.794–1.000)	0.988 (0.964–1.000)

Agreement is generally strong across metrics and recommendation-list lengths, supporting the focus on NDCG@10. Full numerical results are available in the separate `results_reference.pdf` document.